

Task 5: Root-Cause Investigation (SQL Diagnostics)

Objective:

To query the trusted profitability model to isolate the temporal, operational, and commercial drivers of the negative contribution margins, specifically testing for latency spikes and route mix-shifts.

Part 1: Diagnostic SQL & Python Script

This script executes three targeted SQL queries against the modeled data to track margin deterioration, route performance, and the impact of merchant pricing plans.

```
import sqlite3
import pandas as pd

# Connect to the database and load the modeled data
conn = sqlite3.connect('astrapay.db')
df_model = pd.read_csv("trusted_profitability_model.csv")
df_model.to_sql("trusted_profitability_model", conn, if_exists='replace', index=False)

print("--- TASK 5: ROOT-CAUSE INVESTIGATION VERIFICATION ---\n")

# Query 1: Overall Temporal Deterioration (Latency vs. Margin)
query_1 = """
SELECT
    txn_month,
    COUNT(transaction_id) AS success_volume,
    ROUND(AVG(contribution_profit_usd), 2) AS avg_profit_per_txn,
    ROUND(AVG(total_processing_ms), 0) AS avg_processing_ms
FROM trusted_profitability_model
WHERE status = 'SUCCESS'
    AND settlement_fee_usd > 0
GROUP BY txn_month
ORDER BY txn_month;
"""

print("1. Overall Temporal Deterioration:")
print(pd.read_sql_query(query_1, conn).to_string(index=False))
print("\n" + "="*50 + "\n")

# Query 2: Commercial Mix-Shift & Route Deterioration
query_2 = """
SELECT
    txn_month,
    route_id,
    COUNT(transaction_id) AS volume,
    ROUND(AVG(contribution_profit_usd), 2) AS avg_profit
```

```

FROM trusted_profitability_model
WHERE status = 'SUCCESS'
GROUP BY
    txn_month,
    route_id
ORDER BY
    route_id,
    txn_month;
"""

print("2. Route Deterioration & Mix-Shift:")
print(pd.read_sql_query(query_2, conn).to_string(index=False))
print("\n" + "="*50 + "\n")

```

```

# Query 3: Merchant Pricing Impact
query_3 = """
SELECT
    txn_month,
    pricing_plan,
    COUNT(transaction_id) AS volume,
    ROUND(AVG(contribution_profit_usd), 2) AS avg_profit
FROM trusted_profitability_model
WHERE status = 'SUCCESS'
GROUP BY
    txn_month,
    pricing_plan
ORDER BY
    avg_profit ASC
LIMIT 5;
"""

print("3. Merchant Pricing Impact (Most Unprofitable Segments):")
print(pd.read_sql_query(query_3, conn).to_string(index=False))

```

Part 2: Diagnostic Results

Output confirming the operational latency spike in February and the commercial mix-shift penalty in March:

--- TASK 5: ROOT-CAUSE INVESTIGATION VERIFICATION ---

1. Overall Temporal Deterioration:

txn_month	success_volume	avg_profit_per_txn	avg_processing_ms
2026-01	42	-2.60	899
2026-02	38	-16.88	3272
2026-03	45	-21.89	915

=====

2. Route Deterioration & Mix-Shift (Excerpt):

txn_month route_id volume avg_profit

2026-01	R_IN_01	28	2.10	<-- Profitable primary route active
2026-02	R_IN_01	12	-4.50	<-- Primary route degraded (latency)
2026-03	R_IN_02	35	-31.40	<-- Expensive backup route takes over

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3. Merchant Pricing Impact (Most Unprofitable Segments):

txn_month pricing_plan volume avg_profit

2026-03	STANDARD_2.0%	24	-42.15	<-- Fixed pricing fails on expensive routes
2026-02	STANDARD_2.0%	19	-22.30	
2026-03	CUSTOM_TIER	8	-18.45	